RETAIL ANALYTICS · CUSTOMER SEGMENTATION

One in five customers brings in three-quarters of the revenue. *£3.29M of lifetime value is drifting away.*

· Julian Batto-Hokson · UCI Online Retail II · 1.07M invoice lines · Dec 2009 to Dec 2011

· Python · SQL · dbt · DuckDB · scikit-learn · CrewAI · October 2026

Scope note: this is real transaction data from a UK online giftware retailer, many of whose customers are wholesalers. 21.8% of invoice lines have no customer ID (mostly guest checkouts), so the segments describe registered customers, and revenue figures here are registered-customer revenue, not total company revenue.

EXECUTIVE SUMMARY

I cleaned and loaded 1.07M raw invoice lines into DuckDB, modeled them with dbt into a star schema, and segmented the 5,832 active registered customers on recency, frequency and spend (RFM) with K-Means. A local AI agent drafts retention emails for the segments at risk, and a weekly GitHub Actions job rebuilds everything and emails an alert if a check fails.

Revenue is concentrated: 1,189 "Loyal Champions" (20% of customers) account for 73.6% of net revenue. Two segments are slipping away: 1,440 At-Risk Whales (median 184.5 days since their last order, £2.68M lifetime) and 1,950 Dormant Buyers (median 402 days, £0.61M).

Champions lapse fast. If no one ordered for 90 days, the trained model would re-score 537 of the 1,189 Champions as At-Risk Whales. A retention program has to watch Champions' recency, not just customers who have already gone quiet.

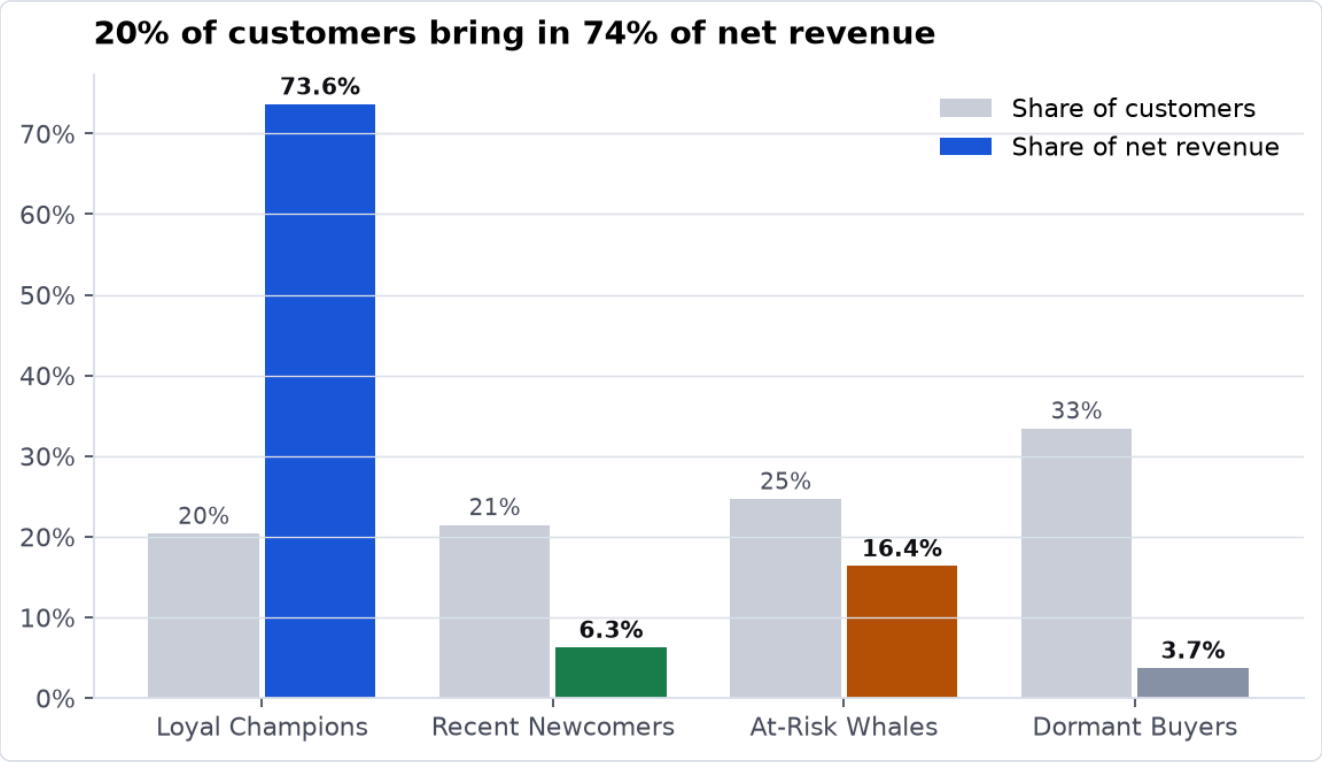
5,832	73.6%	£3.29M	537
Active registered customers segmented	Net revenue from Loyal Champions (20% of customers)	Lifetime value in the two churn-risk segments	Champions re-scored At-Risk after 90 days without orders

-2.0%	0.363
Net revenue, last 12 full months vs prior 12	Silhouette, k = 4 on log-scaled RFM

ANALYSIS

What the data shows

01 Revenue depends on 1,189 customers

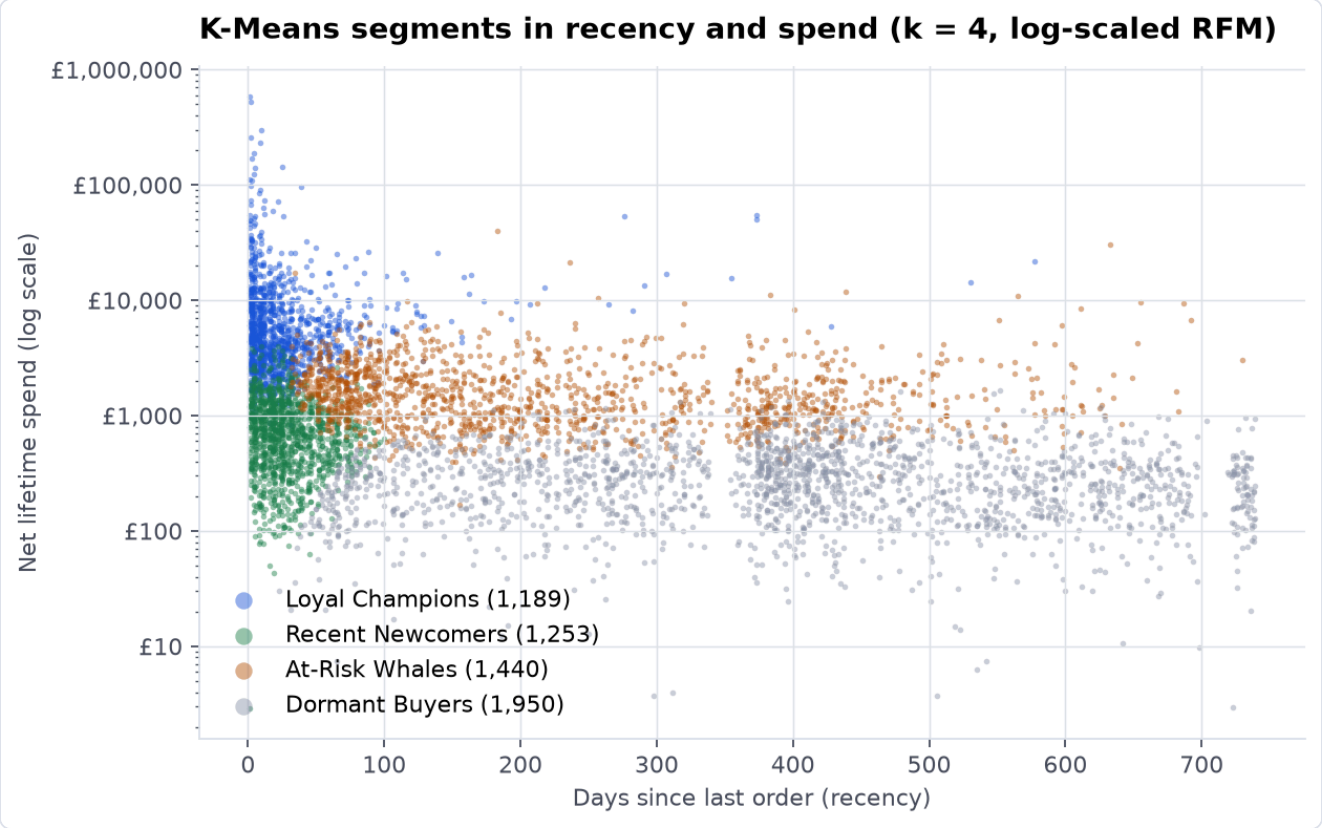


Segment	Customers	Median days since last order	Median orders	Median net spend	Share of net revenue
Loyal Champions	1,189	17	13	£4,809	**73.6%**
Recent Newcomers	1,253	24	3	£705	6.3%
At-Risk Whales	1,440	184.5	4.5	£1,432	**16.4%**
Dormant Buyers	1,950	402	1	£267	**3.7%**

INTERPRETATION

Losing a few dozen Champions costs more than losing every Dormant Buyer. Retention spend should follow value: perks for Champions and Whales, low-cost reactivation for Dormant Buyers.

02 The segments separate cleanly on recency and spend

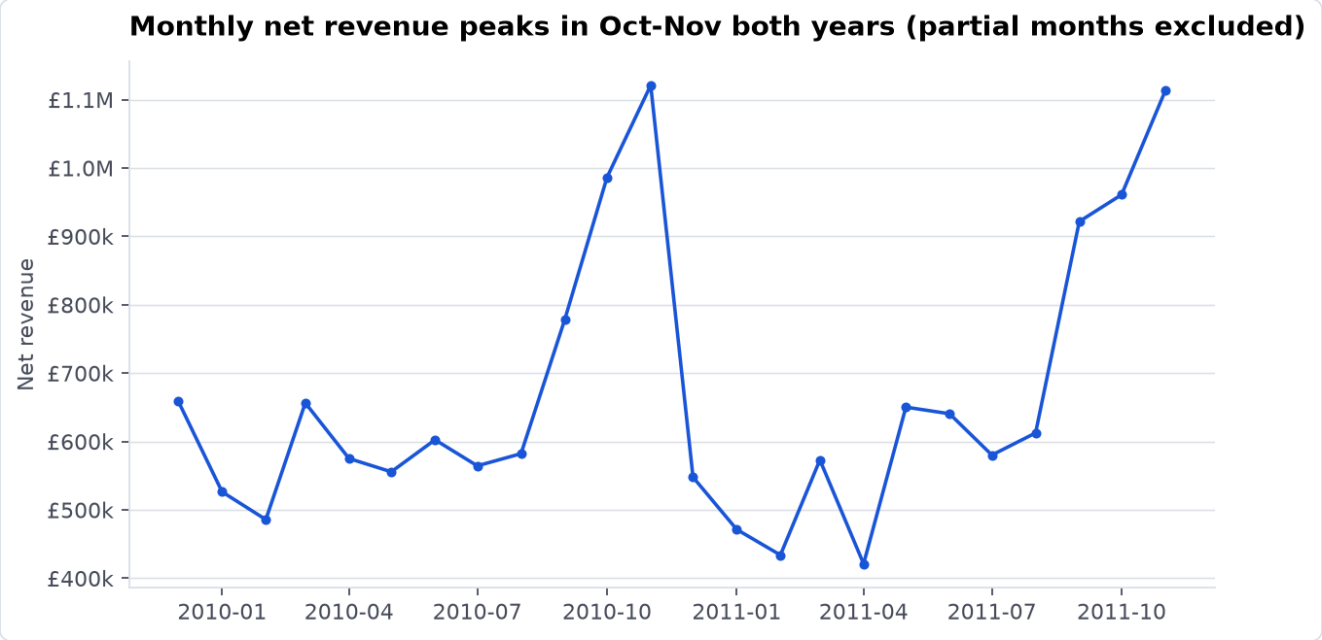


Champions sit top-left (recent, high spend). Newcomers are recent but low spend. Most Whales (the middle 90%) have gone 51 to 513 days without an order, at mid-to-high spend. Dormant Buyers are mostly (69%) one-time, low-spend buyers, with a median of 402 days since their last order.

"WHALE" IS RELATIVE

At-Risk Whales have a median net spend of £1,432. That's above this customer base's median, but most of the very largest accounts are Champions. The persona names describe position, the medians describe size.

03 Revenue is seasonal and slightly down year over year

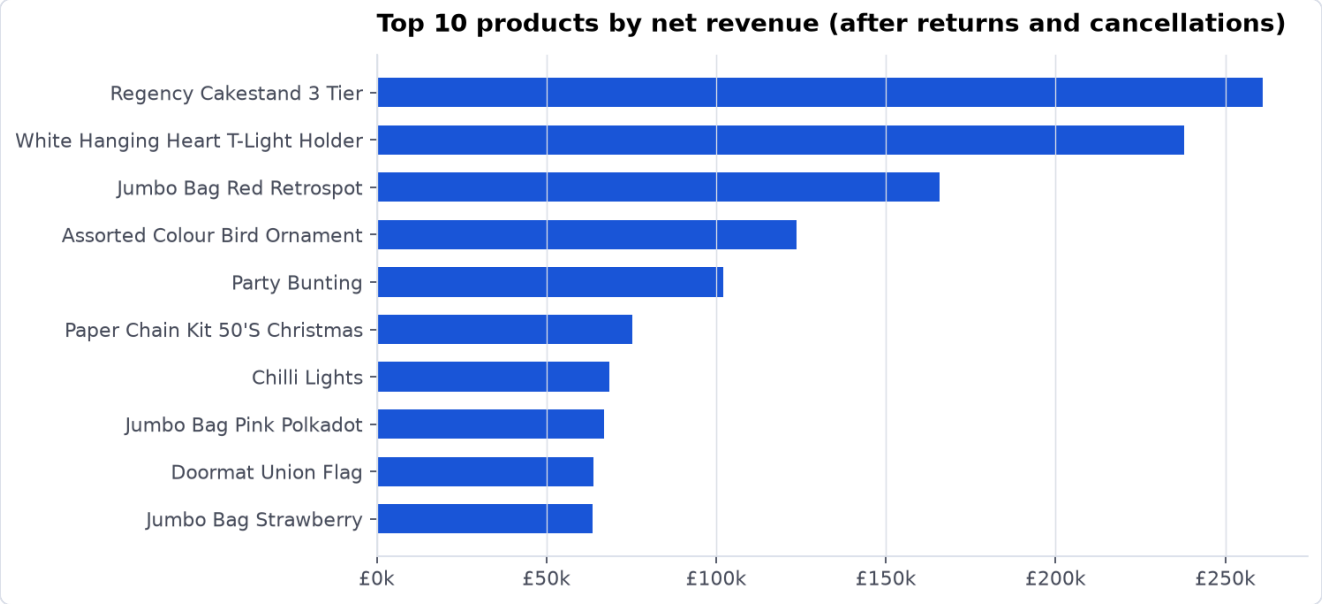


Net revenue peaks in October and November ahead of Christmas in both years (November 2010: £1.12M; November 2011: £1.11M). The last 12 full months (Dec 2010 to Nov 2011) came to £7.93M, down 2.0% from £8.09M in the prior 12. December 2011 is partial (data ends 2011-12-09) and is excluded.

TIMING

Win-back campaigns for lapsed accounts have the most to gain if they land before the September ramp, when buyers plan holiday stock.

04 Best sellers are cheap, high-volume gift items



The Regency cakestand (£261k net, 1,314 customers) and the white hanging heart T-light holder (£238k, 1,490 customers) lead. Products are ranked net of returns: the raw data contains single orders of 80,995 and 74,215 units that were cancelled in full, and ranking by gross revenue would put one of them in the top three.

05 The UK is 84% of revenue, and a handful of overseas accounts are huge

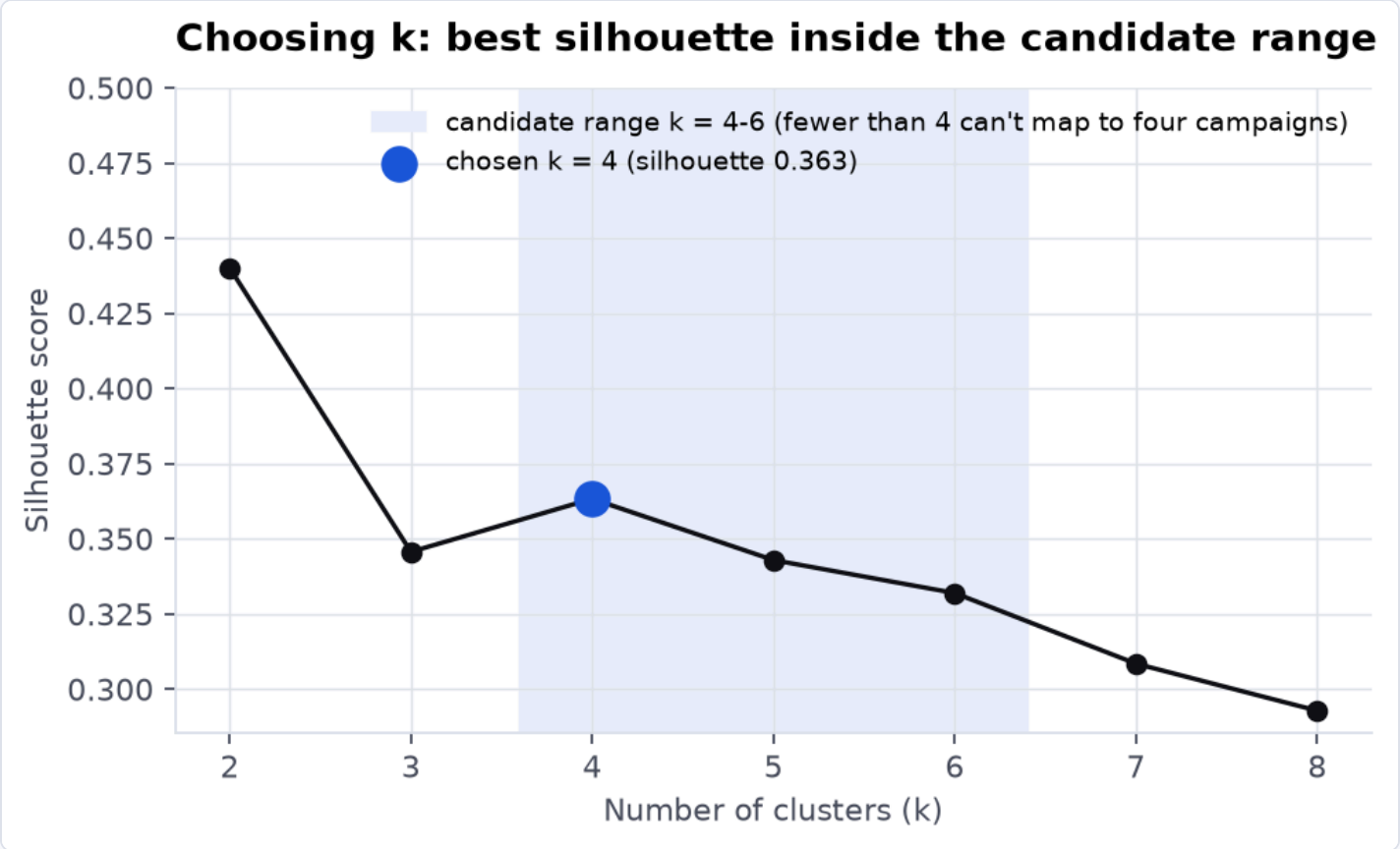
Country	Customers	Net revenue	Share
United Kingdom	5,334	£13.66M	83.5%
EIRE (Ireland)	3	£0.57M	3.5%
Netherlands	22	£0.55M	3.3%
Germany	106	£0.37M	2.3%
France	93	£0.29M	1.8%

Three Irish accounts spend more than 106 German ones combined. These are wholesalers, which is also why spend is so skewed (max £578k against an £840 median) and why the model log-transforms RFM. Full table:

`outputs/revenue_by_country.csv`, 41 countries.

MODEL

Why k = 4, and why the silhouette is "only" 0.363



RFM is log-transformed and standardized, then K-Means picks k from 4 to 6 by silhouette. k = 2 scores higher (0.44) but can't support four distinct campaigns. On raw (unlogged) values K-Means scores 0.59 at k = 4, but only by putting 4 and 37 wholesale accounts in clusters of their own. The log-scale solution gives four balanced clusters (1,189 to 1,950 customers) a marketer can act on. Clusters are named by matching centroids to persona prototypes with the Hungarian algorithm, so names don't depend on K-Means' random cluster IDs.

`k = 4` `silhouette 0.363` `5,832 customers` `3 runs byte-identical`

LOCAL AI

Retention emails drafted by a free local model

A CrewAI agent on Ollama (llama3.2, 3B) drafts a campaign per at-risk segment: strategy, A/B subject lines, email body and follow-up. Facts and copy are kept apart on purpose. The cohort snapshot at the top of each draft is rendered from SQL, so the model can't misquote the numbers, and a guardrail checks structure, the merge field and a per-segment discount cap (15% for Whales, 20% for Dormant) before a draft is accepted.

REAL OUTPUT, UNEDITED

Both at-risk drafts in `campaigns/samples/` passed the guardrail on the first attempt. The Whales draft offers a VIP perk (free shipping on the next 3 orders) with no discount, which is what the policy asks for. The model still slips: it suggests sending "7-10 days after the last order" to customers whose median gap is 184 days, and the Dormant follow-up invents a phone number. That's why every draft is labeled "Review before sending".

NEXT STEPS

What I'd recommend

RETENTION

Trigger an alert when a Champion passes 60 days without an order

Champions' median gap is 17 days. Waiting until they look like Whales means acting after 537 of them have already drifted.

```
Weekly list: Champions with recency
over 60 days
```

WIN-BACK

VIP perks for Whales before discounts

They already spend £1,432 at the median. A perk protects margin; test it against a 10% code with a holdout group.

```
Measure: 90-day reorder rate, perk vs
discount vs holdout
```

DATA

Capture customer IDs at guest checkout

21.8% of lines can't be tied to a customer, so they can't be segmented or retained.

```
Target: under 10% unattributed lines
```

TECHNICAL REFERENCE

Data and pipeline

Dimension	Value
Source	UCI Online Retail II (CC BY 4.0), downloaded at build time
Rows	1,067,371 raw lines → 776,579 clean sale lines and 17,586 returns; every other row kept in an audit table with a reason
Cleaning	9-day overlap between the two source sheets removed (22,523 rows that would double-count), duplicates, bad-debt adjustments, non-product codes, missing IDs
Snapshot	Recency measured from 2011-12-10, the day after the last invoice
Testing	51 dbt checks; 88 pytest tests including 30 health checks on the real warehouse; weekly rebuild in GitHub Actions with an email alert on failure
Tools	Python (pandas, NumPy, scikit-learn, SciPy), SQL, dbt-duckdb, DuckDB, Streamlit, Plotly, Matplotlib, CrewAI, Ollama, GitHub Actions